Multiplying Polynomials — Answer Key

Type 1 Multiplying Polynomials

1. $(x+2)(x+3)$: $x \cdot x + 3 \cdot x + 2 \cdot x + 2 \cdot 3 = x^2 + 3x + 2x + 6$
Answer: $x^2 + 5x + 6$

2. $(x+4)(x-1)$: $x^2 - x + 4x - 4$
Answer: $x^2 + 3x - 4$

3. $(x-5)(x-2)$: $x^2 - 2x - 5x + 10$
Answer: $x^2 - 7x + 10$

4. $(2x+1)(x+3)$: $2x^2 + 6x + x + 3$
Answer: $2x^2 + 7x + 3$

5. $(x+1)(x^2+2x+3)$: $x^3 + 2x^2 + 3x + x^2 + 2x + 3$
Answer: $x^3 + 3x^2 + 5x + 3$

Type 2 Square of a Binomial: $(a+b)^2$ and $(a-b)^2$

6. $(x+3)^2$: $x^2 + 2 \cdot x \cdot 3 + 3^2$
Answer: $x^2 + 6x + 9$

7. $(x-4)^2$: $x^2 - 2 \cdot x \cdot 4 + 4^2$
Answer: $x^2 - 8x + 16$

8. $(x+5)^2$: $x^2 + 2 \cdot x \cdot 5 + 5^2$
Answer: $x^2 + 10x + 25$

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9. $(2x+1)^2$: $(2x)^2 + 2 \cdot (2x) \cdot 1 + 1^2$
Answer: $4x^2 + 4x + 1$

10. $(x-6)^2$: $x^2 - 2 \cdot x \cdot 6 + 6^2$
Answer: $x^2 - 12x + 36$

Type 3 Sum Times Difference: $(a+b)(a-b)$

11. $(x+5)(x-5)$: $x^2 - 5^2$
Answer: $x^2 - 25$

12. $(x-7)(x+7)$: $x^2 - 7^2$
Answer: $x^2 - 49$

13. $(2x+3)(2x-3)$: $(2x)^2 - 3^2$
Answer: $4x^2 - 9$

14. $(x+10)(x-10)$: $x^2 - 10^2$
Answer: $x^2 - 100$

15. $(3x-1)(3x+1)$: $(3x)^2 - 1^2$
Answer: $9x^2 - 1$

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